tions. As compared to the Cortex A75, the Cortex A76 claims 40 percent better energy efficiency, and 35 percent better performance. Now let's take a look at the block diagram of the Mongoose 4 core used in the Exynos 9820 chip.

Generally, a core is divided into three parts, the frontend, the execution engine and the memory cache. A predictor interprets the instructions from the software in an instruction cache, and sends it along to the instruction pipeline. From here, a microsequencer may decode the bits in the instruction, and select microinstructions to perform. In any case, the instructions are then sent on to a buffer in the execution engine, and then to a dispatch queue. From here, the instructions are piped to an integer cluster which has a complex of address generation units and arithmetic logic units, or to a floating point unit, depending on the instruction. The processed data then moves to a data cache, then to the level 2 cache, and out of the core. The interface of the core has a bandwidth of so many bytes per clock cycle.

Now the amount of L1 cache is usually fixed by the core. L2 cache can support a range of values, say 256KB to 512KB. The L3 cache is usually optional, but can support even more memory, which can even go up to 4MB. All of the Cortex cores are available with optional cryptography units.

THE GPUS

The Kirin 980 and the Exynos 9820 both have use the Mali G76 GPU. The Helio P90 has the rather odd choice of a PowerVR GM 9446 GPU, a company that used to make just about the best GPUs for mobile devices a few years ago. The Snapdragon 855 uses an Adreno 640 GPU, which is not surprising considering that Adreno is an anagram of Radeon, a rebranding that occurred after Qualcomm acquired handheld and multimedia assets from AMD back in 2009, just when smartphones were about to take over the world. Apple as usual, uses a GPU of its own design. Qualcomm with Adreno and ARM with Mali are the two most widely used smartphone GPU suppliers worldwide.

[JARGON BUSTER]

GHz
GigaHertz, A common measure for the clock speed of a processor. Depending on the microarchitecture, different processors can take a different number of clock cycles to execute a particular type of operation.

Heatsink
A heatsink is a device that keeps processors and other components cool. Heatsinks can be active, or passive, or incorporate both active and passive components.

Hyperthreading
Technology developed by Intel that allows the same processor to simultaneously process two instruction sets, converting a single processor into two virtual processors. Software written for multiple processor systems can easily take advantage of hyperthreading.

ILP
Instruction level parallelism is a measure of how many parallel processes can be simultaneously handled by a computer, and is important for graphically intensive as well as scientific applications.

Instruction Set
An instruction set is an abstract model for a computer, that allows developers to write code for the machine. It is an interface between the software and the hardware.

Instructions
Instructions are steps that a computer should take to solve a particular problem.

The latest offerings by Mali, Adreno and PowerVR all support the most used APIs, OpenCL, OpenGL and DirectX 12 while PowerVR also has support for the Android NN HAL API.

Now the specific implementation of the GPU also matters, and this depends on the manufacturer. For example, the Mali GPU supports scaling, and offers a choice of between 4 to 20 GPU cores. There is also some freedom in how the cache is configured, in slices of 2 or 4 memory units. In Adreno, Qualcomm supports Physically Based Rendering, or PBR, which is essentially a shader that provides more realistic images and accurate lighting. While the Mali G76 is made with the 7nm process, the Adreno 640 uses a 10nm process. The power consumption of the Mali GPUs should be lower than the Adreno GPU at comparable performance levels.

The latest generation of GPUs are designed to support HD video recording, HD video playback, HDR video in some cases, 360 degree video playback, as well as VR/AR applications. The GPU architectures are ideally suited to take the workload away from CPUs for some applications, and in smartphones, the latest generation of GPUs have features oriented towards support for machine learning applications.

WHAT'S NEXT

Well, expect the miniaturisation to continue. TSMC has just announced that it has completed designs for it's 5nm node, which will be used for the next generation of SoCs. These will be geared towards mobile computing with a heavier dependence on AI and 5G, and lower power requirements. It is not clear at this point of time which company will be the first to use the new node, but Apple has been at the forefront for the past few years, and was the first to use the 7nm node. Eventually, all manufacturers can be expected to jump to this node in the early 2020s. The 5nm node offers a 45 percent reduction in area over the 7nm node, but only a 15 percent improvement in terms of energy requirements. At this scale, these are quite significant power savings and help with increasing power efficiency. d